

4_numeric_data_types_and_strings

```
a=2  
print(a," is of type ", type(a))
```

[1] ✓ 0.0s

... 2 is of type <class 'int'>

```
a=2.5  
print(a," is of type ", type(a))
```

[2] ✓ 0.0s

... 2.5 is of type <class 'float'>

▷ ▾

```
a=2+3j  
print(a," is of type ", type(a))  
print("a is complex number? ", isinstance(2+3j,complex))
```

[3] ✓ 0.0s

... (2+3j) is of type <class 'complex'>
a is complex number? True

String Data types

```
single_quote='Hanu'
print(single_quote," is of type ", type(single_quote))
```

[4] ✓ 0.0s

```
... Hanu is of type <class 'str'>
```

```
double_quote="Hanu"
print(double_quote," is of type ", type(double_quote))
```

[5] ✓ 0.0s

```
... Hanu is of type <class 'str'>
```

```
triple_quote="""Hanu
rao
allada"""
print(triple_quote," is of type ", type(triple_quote))
```

[8] ✓ 0.0s

```
... Hanu
rao
allada is of type <class 'str'>
```

Indexing and Slicing

indexing for a given string starts from zero

```
mystring='python'
print(mystring[0]) # prints first character
print(mystring[1]) # prints second character
print(mystring[1:4]) # prints characters starting from 2nd to 4th
print(mystring[2:]) # prints characters starting from 3rd character
```

[9] ✓ 0.0s

```
... p
y
yth
thon
```

If we refer to the index position which is greater than the length of the string then we will get **index array out of range error**

String Concatenation

```
string1='Hello'
string2='World'
print(string1 + ' ' + string2) # prints Hello World
concat_string=string1 + ' ' + string2
print(concat_string) # prints Hello World
```

[10] ✓ 0.0s

... Hello World
Hello World

```
string3=10
concat_string=string1 + ' ' + string2 + ' ' + string3
print(concat_string)
```

[12] ✗ 0.1s

TypeError Traceback (most recent call last)
/tmp/ipykernel_26650/308332608.py in <cell line: 0>()
 1 string3=10
----> 2 concat_string=string1 + ' ' + string2 + ' ' + string3
 3 print(concat_string)

TypeError: can only concatenate str (not "int") to str

```
string3=10
concat_string=string1 + ' ' + string2 + ' ' + str(string3)
print(concat_string)
```

[13] ✓ 0.0s

... Hello World 10

Common String Functions

- `len()`: Returns length of a string
- `lower()`, `upper()`: converts a string to lower or upper case
- `strip()`: removes leading and trailing spaces
- `replace()`: Replaces a substring with another substring
- `split()`: Splits a string into list of substrings based on delimiter
- `find()`: `Index()`: Finds the index of substring in the string



```
mystring="      Hello, World      "
print(len(mystring)) # prints 27
print(mystring.strip()) # prints Hello World
print(len(mystring.strip())) # prints Hello World
print(mystring.lower())
print(mystring.upper())
print(mystring.replace("Hello","Hi"))
print(mystring.replace("o","*"))
print(mystring.strip())
print(mystring.split(","))
print(mystring.split(" "))
```

[18] ✓ 0.0s

... 31

Hello, World

12

hello, world

HELLO, WORLD

Hi, World

Hell*, W*rld

Hello, World

[' Hello', ' World ']

['', '', '', '', '', '', '', '', 'Hello,', 'World', '', '', '', '', '', '', '', '', '']